Introduction to Biomaterials

Lecture [3]

Biology and Materials Science: Basic Principles

Goals

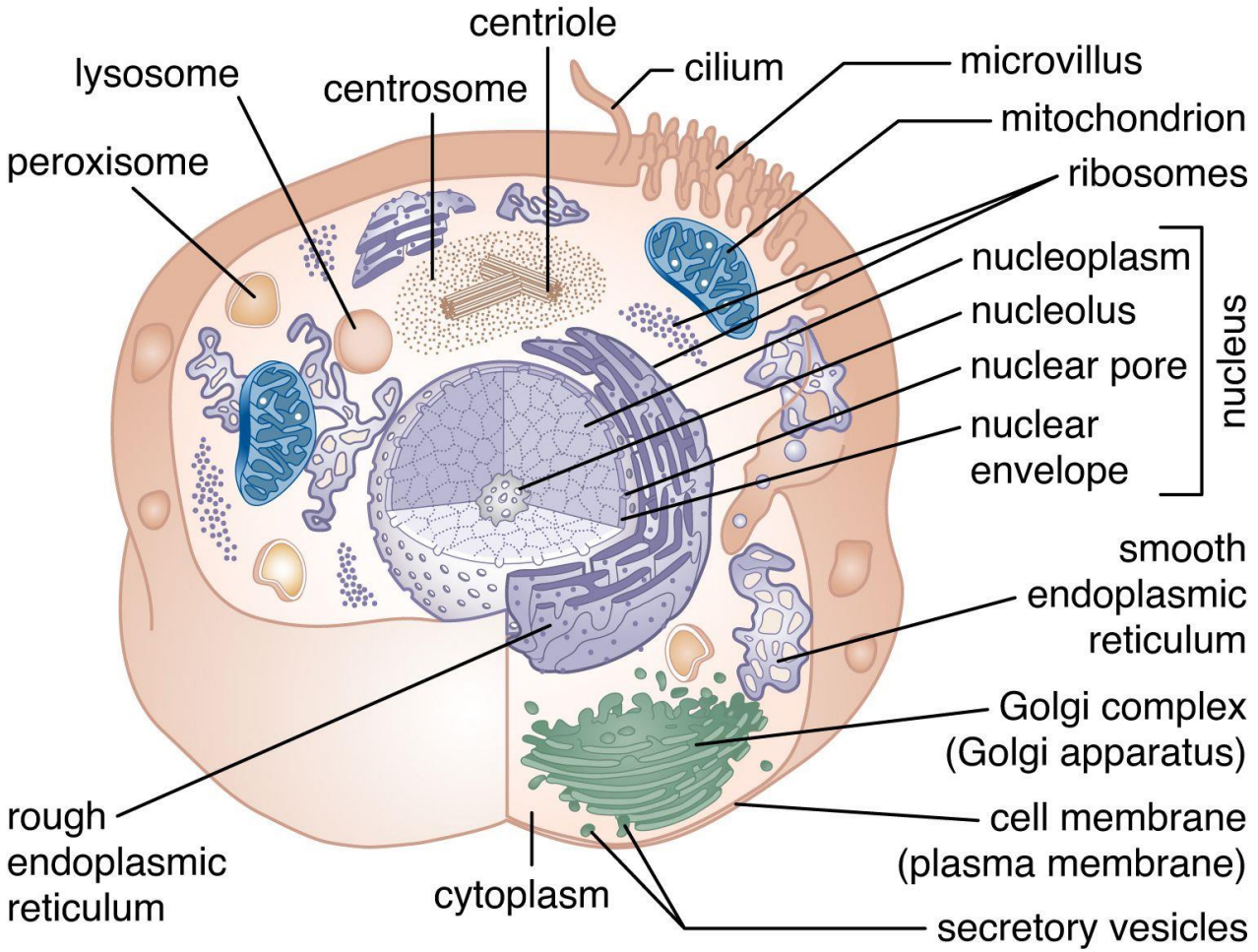
- **After reading this chapter, students will understand the following:**
  - Cell structure and body response to materials
  - Transport: flow, flux, diffusion
  - Cell–material interaction and healing
  - Design of safe and functional biomaterials

Basic Cell Structure (for Biomaterials)

- Cells are the main living units that interact with biomaterials. Important structures include:
  - **Cell membrane:** controls transport and senses surfaces
  - **Nucleus:** regulates genes and cell response
  - **Mitochondria:** provide energy (ATP)
  - **Cytoskeleton:** controls shape and attachment
  - **Receptors:** allow communication with environment

Basic Cell Structure

- Why it matters in biomaterials:
  - Biomaterials influence cells mainly through the membrane, receptors, and mechanical signaling, which can change adhesion, growth, healing, or inflammation.



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The Biological Environment

- Biological systems are complex, but in biomaterials engineering we study them at different levels—molecules, proteins, cells, tissues, and the whole body—to understand how a material will behave inside the body.
- This helps us predict cell response, healing, integration, and possible failure of implants.

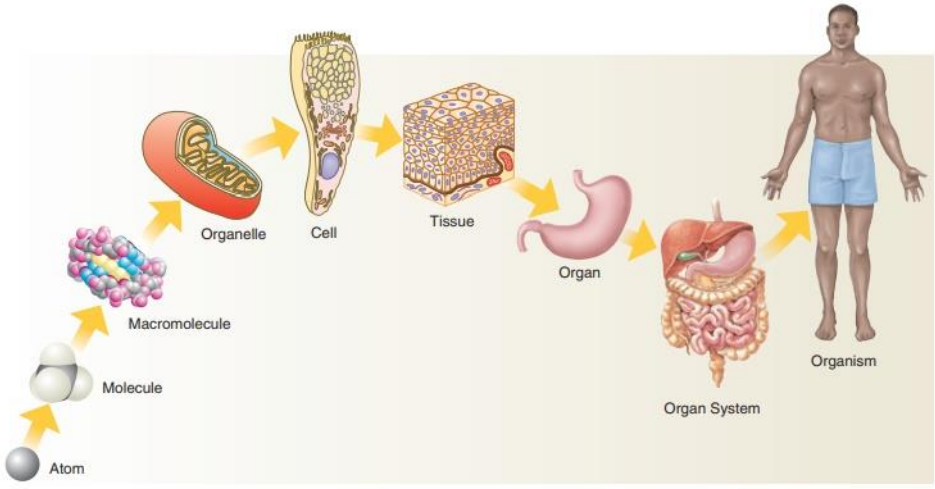


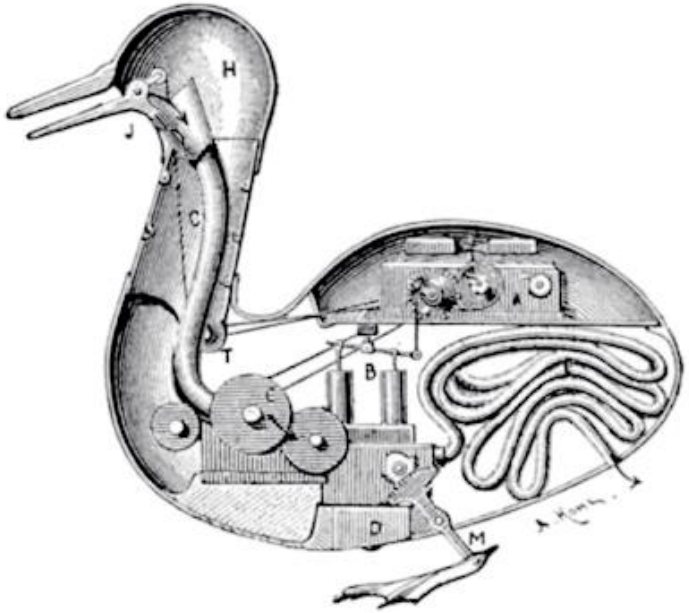
FIGURE 1-1 Organization levels of the body.

Adapted from Shier, D.N., Butler, J.L., and Lewis, R. Hole's Essentials of Human Anatomy & Physiology, Tenth edition. McGraw Hill Higher Education, 2009.

Reductionism in Biomaterials

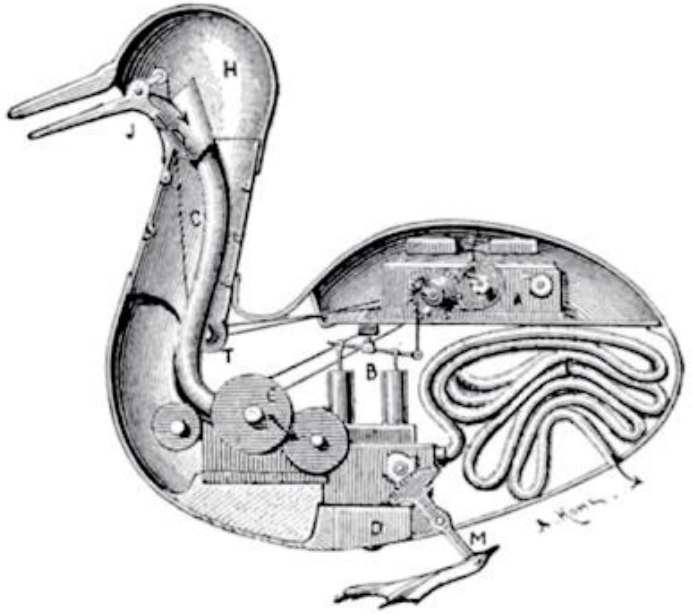
> *It can scarcely be denied that the supreme goal of all theory is to make the irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience.* — *A. Einstein, 1933*

http://biofisica.info/articles-15/we-are-reductionists-should-we-care/



Reductionism in Biomaterials

- Biological systems are often studied by breaking them into smaller parts such as cells, proteins, and biochemical signals.
- This approach is useful in biomaterials because engineers often test materials first at the protein, cell, or tissue level before clinical use.
- It helps explain how a material surface affects cell attachment, inflammation, or protein adsorption.

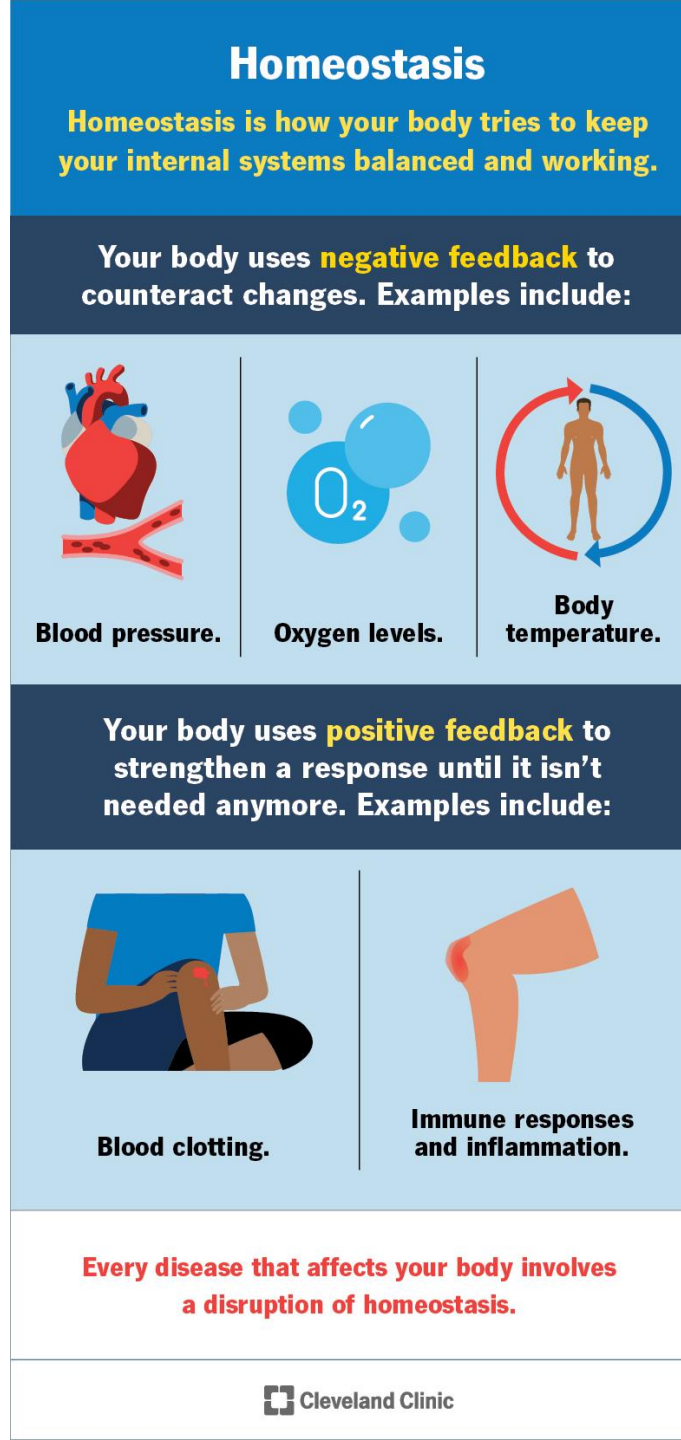


Reductionism in Biomaterials

- However, simplified tests such as in vitro experiments may not fully represent the real body environment.
- A material that looks good in the lab may behave differently in vivo because the body has more complex signaling, immune response, and hormonal control.
- Why it matters in biomaterials:
  - Engineers must be careful when interpreting lab results.
  - Biomaterials should be evaluated with awareness that the real biological environment is more complex than isolated experiments.

Homeostasis

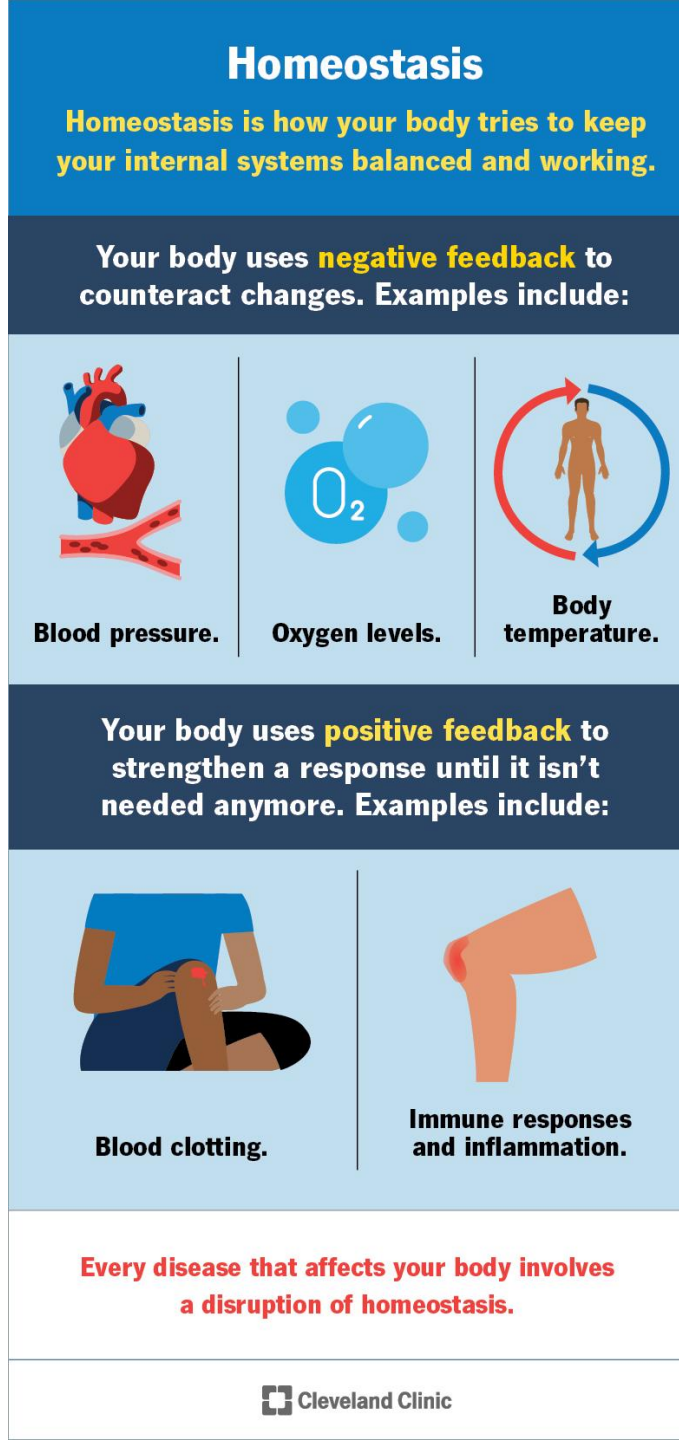
- Living systems try to maintain a stable internal condition called homeostasis.
- Cells need a controlled environment with proper nutrients, oxygen, ions, and waste removal to survive and function.
- Any implanted biomaterial enters this balanced system and may disturb it.



Khalid Abushahla, Doctoral Researcher in Biomedical Engineering

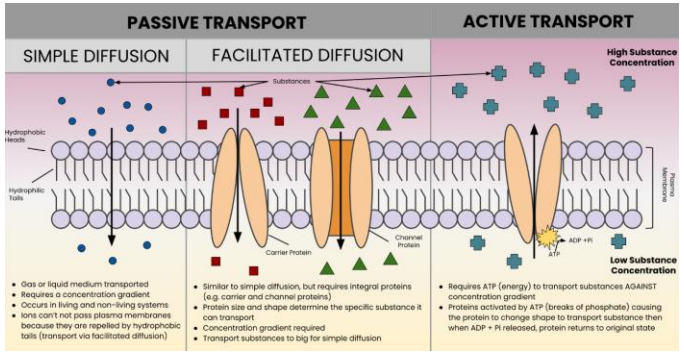
Homeostasis

- Why it matters in biomaterials:
  - A biomaterial should support or at least not strongly disrupt the local biological balance.
  - Good implant design tries to preserve normal tissue conditions and reduce harmful responses such as inflammation, toxicity, or transport blockage.



Transport in Biological Systems

- Cells constantly exchange substances with their surroundings.
- Nutrients move toward cells, while waste products move away from them.
- This transport occurs across structures such as:
  - cell membranes
  - interstitial fluid
  - capillary walls
  - blood plasma
- Transport may occur by passive or active mechanisms.

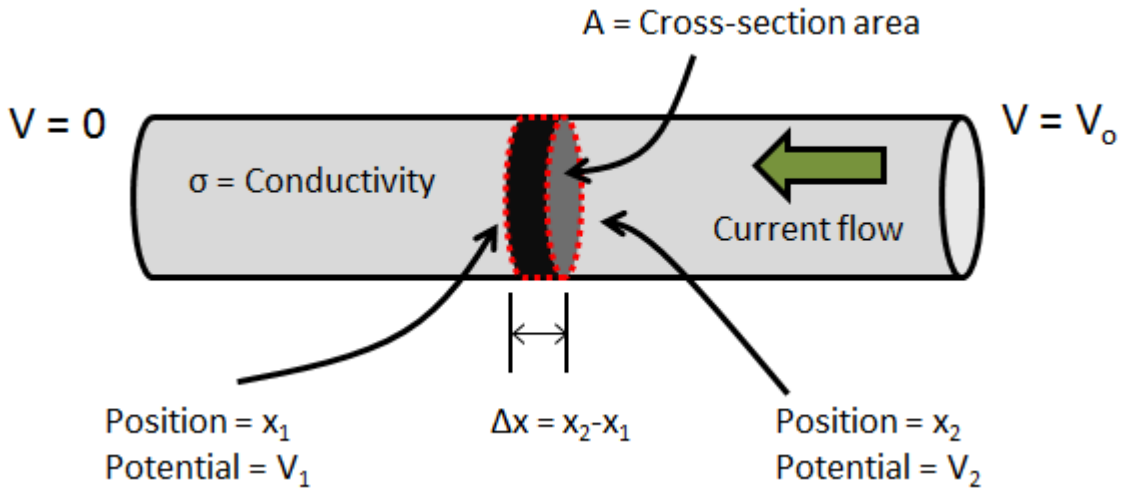


Transport in Biological Systems

- Why it matters in biomaterials:
- Biomaterials often depend on transport processes for success.
- For example, transport controls:
  - oxygen and nutrient delivery near an implant
  - drug release from a biomaterial
  - ion release from a surface
  - protein penetration into scaffolds or hydrogels
  - degradation products leaving biodegradable materials

Flow and Flux

- Transport can be described as flow or flux.
- **Flow** means how much material moves per unit time.
- **Flux** means flow normalized by area, so it describes transport more precisely across a surface or membrane.



Flow and Flux

- Simple idea:
  - Flow = how much moves.
  - Flux = how much moves through each unit area.
- Example in biomaterials:
  If oxygen passes through an implant membrane:
  - Flow = total oxygen crossing each second.
  - Flux = oxygen crossing each square meter of membrane each second.

Term	Meaning	Unit Example	Why Important
Flow	Total amount of substance moving per unit time	mL/s, mol/s	Measures overall transport
Flux	Flow per unit area	$mol/m^2 \cdot s$	Measures transport through a surface

Flow and Flux

- Why it matters in biomaterials:
  - Flux is very important because biomaterials interact with the body mainly through their surface.
  - Engineers use flux concepts to study how substances move across implant surfaces, membranes, coatings, and tissue-engineered scaffolds.

Diffusion and Concentration Gradients

- One major transport mechanism is diffusion, where substances move from high concentration to low concentration.
- This is described by Fick's law, which links transport rate to the concentration gradient.
- Diffusion is one of the most basic physical principles used to understand movement in biological systems.

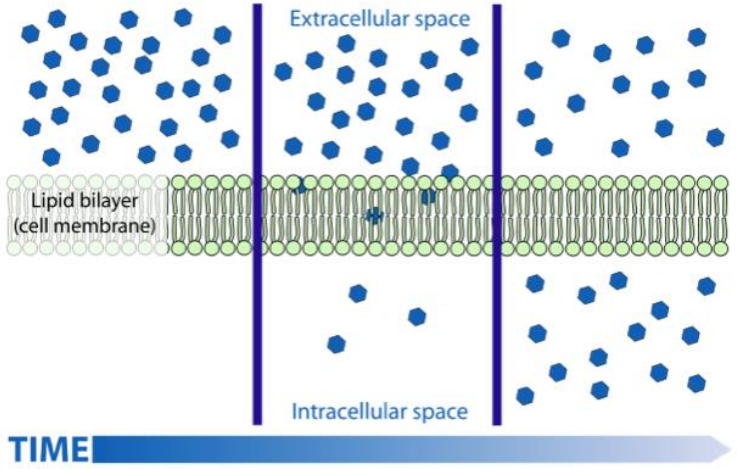


Fig 1. Diffusion Across a Semipermeable Membrane. Molecules of dye travel through solution across the semipermeable membrane from a region of high concentration to a region of low concentration. This occurs until both sides of the semipermeable membrane have an equal concentration of the molecule of dye and a dynamic equilibrium is established.

Fick's First Law:

Term	Meaning	Simple Idea
Concentration Gradient	Difference in concentration between two regions	Substances move from crowded area to less crowded area
Diffusion	Natural movement of particles from high concentration to low concentration	Like perfume spreading in a room
Fick's Law	Describes how fast diffusion happens	Bigger gradient = faster movement

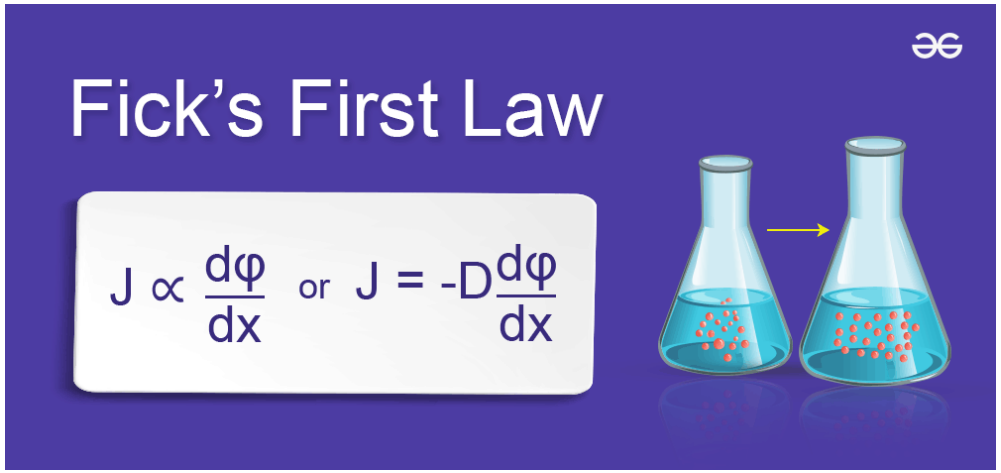
Fick's First Law:

$$J = -D\frac{dC}{dx}$$

Symbol	Meaning
J	Flux (amount moving through area)
D	Diffusion coefficient (how easily substance moves)
dC/dx	Concentration gradient

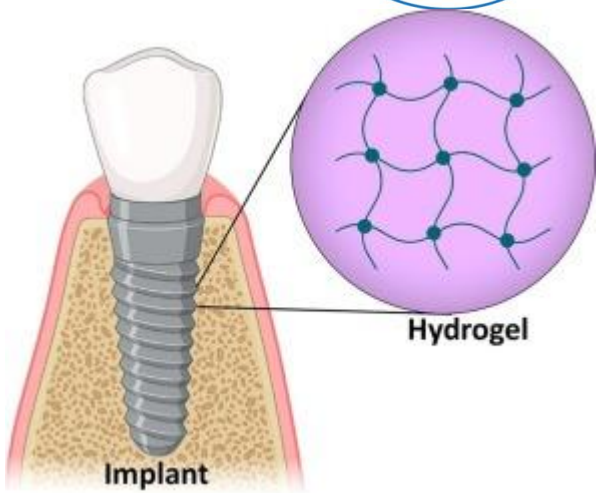
Examples - Fick's First Law:

- **Simple Meaning:**
- If concentration difference is **large** → diffusion is **fast**
- If concentration difference is **small** → diffusion is **slow**
- If material blocks movement → diffusion is slower



Fick's First Law:

- Example:
  - Oxygen outside a cell is high, inside cell is low.
    So oxygen moves into the cell by diffusion.
- Biomaterials Example:
  - In a hydrogel drug implant:
    Drug concentration inside implant is high, outside tissue is low.
    So drug diffuses outward slowly to the body.
- So:
  - High → Low concentration
    Bigger difference = faster diffusion

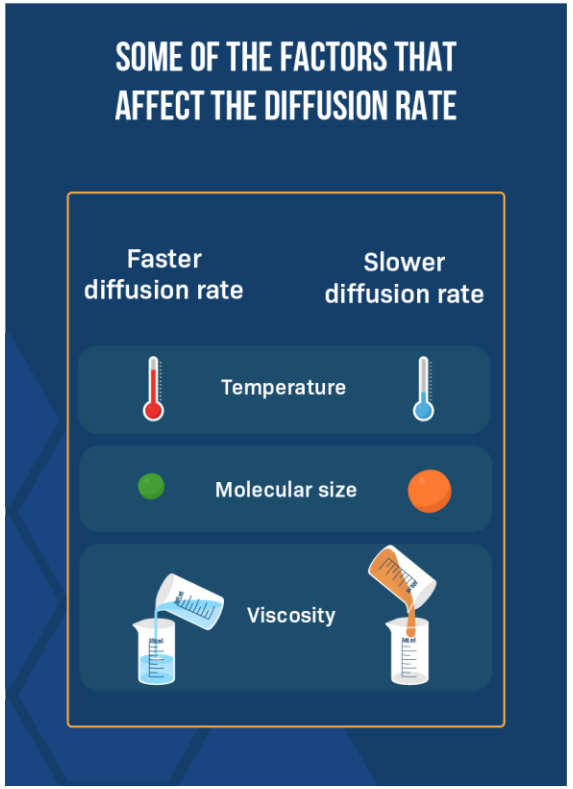


Diffusion and Concentration Gradients

- Why it matters in biomaterials:
  - Diffusion helps explain:
    - drug release from delivery systems
    - movement of ions from bioactive materials
    - transport of proteins into hydrogels
    - exchange of molecules through membranes
  - It is essential in designing biomaterials that must interact with surrounding tissue in a controlled way.

Factors Affecting Diffusion

- Diffusion depends on material and environmental properties such as:
  - Temperature
  - fluid viscosity
  - particle size
- Small particles generally diffuse more easily than large ones.

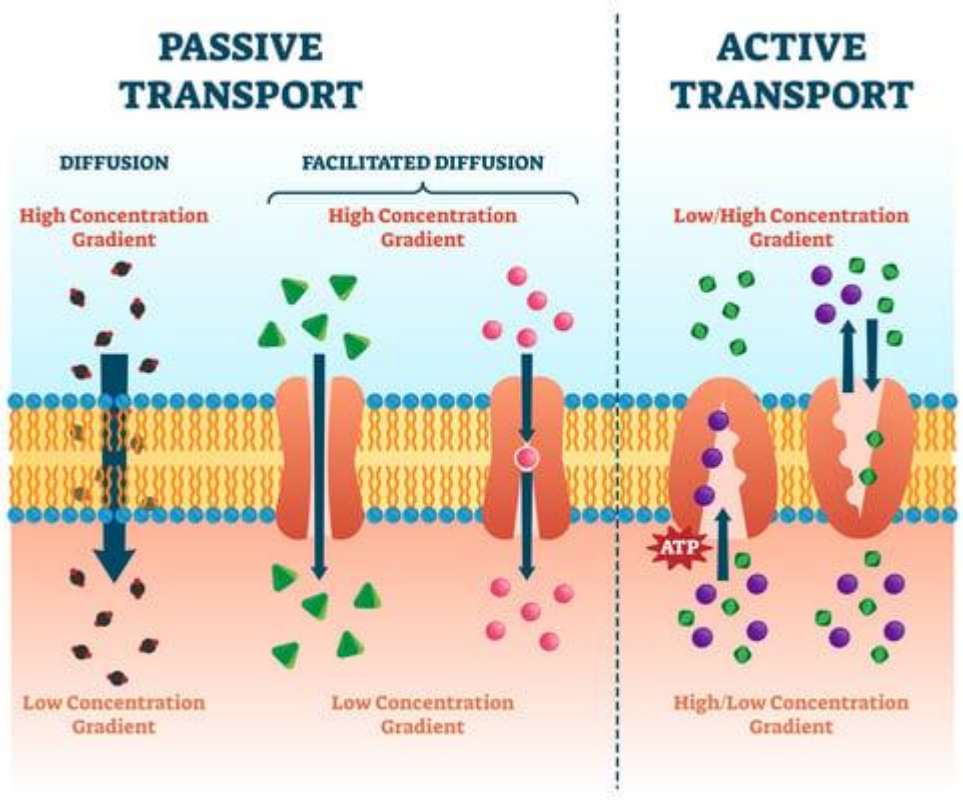


Factors Affecting Diffusion

- Why it matters in biomaterials:
  - These factors affect how fast molecules move near or through a biomaterial.
  - Engineers must consider them when designing:
    - drug carriers
    - porous scaffolds
    - Membranes
    - Hydrogels
    - biodegradable implants

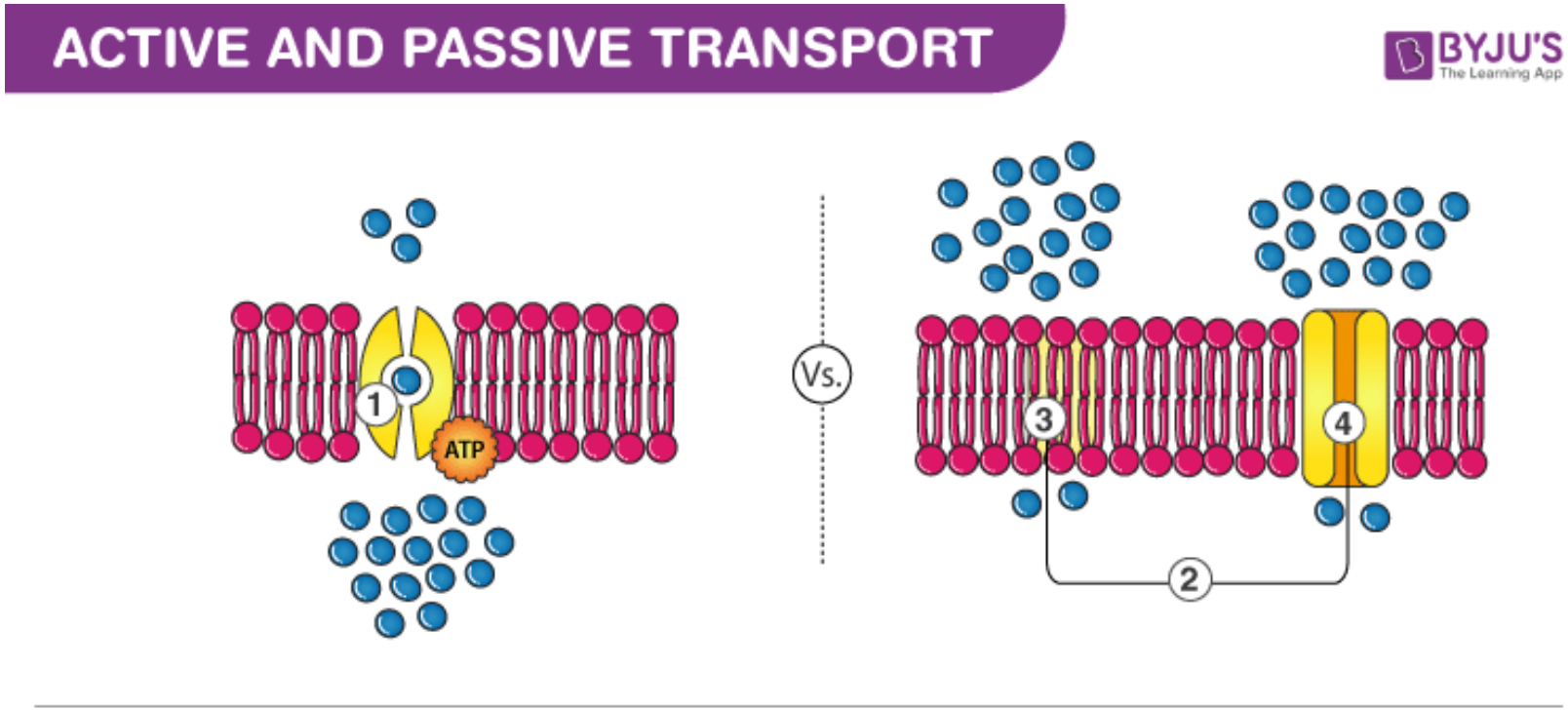
Passive and Active Transport

- Not all biological transport happens passively.
- Some substances move using active cellular mechanisms, which require biological energy and regulation.
- Therefore, real biological transport is more diffusion alone.



Active = needs ATP + goes uphill
Passive = no ATP + goes downhill

- https://byjus.com/biology/difference-between-active-and-passive-transport/



Feature	Active Transport	Passive Transport
Energy Requirement	Requires cellular energy (ATP)	Does not require cellular energy
Direction of Movement	Low concentration → High concentration	High concentration → Low concentration
Against / With Gradient	Moves **against** concentration gradient	Moves **with** concentration gradient
Transported Materials	Ions, proteins, large molecules, complex sugars	Oxygen, water, CO_2, lipids, small soluble molecules
Purpose in Cell	Moves needed substances into/out of cell actively	Helps maintain equilibrium/homeostasis
Nature of Process	Biological / metabolic process	Physical process
Selectivity	Highly selective	Selective or partly non-selective
Speed	Usually faster and regulated	Usually slower (depends on gradient)
Directionality	Often one main direction	Can occur both directions depending on gradient
Carrier Proteins	Usually requires pumps/carrier proteins	Simple diffusion: no protein; Facilitated diffusion: uses proteins
Effect of Oxygen Level	Reduced if cell energy decreases	Not directly dependent on oxygen
Effect of Metabolic Inhibitors	Can stop or reduce process	Usually not affected directly
Examples	Sodium-potassium pump, endocytosis, exocytosis	Diffusion, osmosis, facilitated diffusion

Passive and Active Transport

- Why it matters in biomaterials:
  - Engineers should **not** assume that all cell–material interactions are controlled only by passive physics.
  - Cells actively sense, respond to, and communicate with biomaterial surfaces.
  - This is important in areas such as tissue engineering, implant integration, and cell-responsive materials.

Genetic Regulation and Control Systems

- Cells may have the same DNA, but they behave differently because different genes are turned on or off.
- Phenotype = how the cell looks/functions.
- Genotype = the genetic information inside the cell.
- Cell behavior changes when it receives environmental signals (chemical, physical, mechanical).
- These signals regulate protein production through gene expression.
- https://www.youtube.com/watch?v=4eK2JBctUa0

Genotype vs Phenotype

GENOTYPE	PHENOTYPE	
The genotype is an organism's genetic information.	The phenotype is the set of observable physical traits.	
BB homozygous dominant	purple	
Bb heterozygous	purple	
bb homozygous recessive	white	

sciencenotes.org

Genetic Regulation and Control Systems

- Biomaterials relevance:
  - Biomaterials can influence cell behavior by their surface chemistry, stiffness, and structure.
  - This is important in tissue engineering and regenerative medicine, where materials are designed to guide cell function.

Protein Adsorption

- When a biomaterial is placed inside the body, proteins from blood and body fluids rapidly attach to its surface within seconds or minutes.
- This happens before cells directly contact the material.

- Why it matters in biomaterials:
  - The adsorbed protein layer controls how cells “see” the material. It can influence cell attachment, healing, clotting, inflammation, and implant acceptance.
  - Surface chemistry and roughness strongly affect protein adsorption.

Inflammation and Immune Response

- The body may recognize an implanted material as foreign and activate immune defenses. White blood cells and inflammatory signals may gather around the implant area.

Why it matters in biomaterials:

- A mild short-term response may help healing, but excessive or chronic inflammation can damage tissue, cause pain, fibrosis, loosening, or implant failure.
- Good biomaterials aim to minimize harmful immune reactions.

Cell Adhesion and Integration

- Cells can attach to some material surfaces and begin to grow, spread, and communicate.
- Surface properties such as roughness, stiffness, chemistry, and wettability affect adhesion.

- Why it matters in biomaterials:
  - Good cell adhesion is important for tissue engineering scaffolds, wound healing materials, and bone implants.
  - For example, successful osseointegration of dental or orthopedic implants depends on cell attachment and tissue bonding.

Think with me 🤔🤔🤔

- Define the flow and flux terms and identify the characteristic equation regarding diffusion in biological membranes.



Thank you